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unit 0:

1.) a. $F_{\text{net}} = -0.0332 \text{ N}$

b. $E_{\text{net}} = 2.54 \times 10^7 \text{ N/C}$

2.) a. q is directly downward

b. $F = k \frac{19.921}{r^2} = 76.4 \text{ N downward}$

3.) $KE = eV = (1.6 \times 10^{-19} \text{ C})(40000)$
 $= 6.4 \times 10^{-15} \text{ J}$

$KE = \frac{1}{2} mv^2$

$V = \sqrt{\frac{2KE}{m}}$

$= \sqrt{\frac{2(6.4 \times 10^{-15} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}}$

$= 3.75 \times 10^8 \text{ m/s}$

4.) a. $E = \Delta V/d \rightarrow \Delta V = Ed$
 $= (7.5 \times 10^4)(0.04)$
 $= 3000 \text{ V}$

b.) $V = Ed$
 $= (7.50 \times 10^4)(0.01)$
 $= 750 \text{ V}$

c.) $E = \Delta V/d$
 $= 80 \times 10^{-3} / 9 \times 10^{-9}$
 $= 8.89 \times 10^6 \text{ V/m}$

5.) $E = (32 \times 10^3) \times (1.602 \times 10^{-19})$
 $E = 5.126 \times 10^{-15} \text{ J}$
 $= \frac{5.126 \times 10^{-15} \text{ J}}{2(1.6 \times 10^{-19})(0.02)}$
 $= 8.00 \times 10^5 \text{ V/m}$

6.) $r = \frac{kq_1q_2}{k}$
 $= \frac{(8.99 \times 10^9)(3.204 \times 10^{-19})(1.265 \times 10^{-17})}{8.01 \times 10^{-15}}$
 $r = 4.55 \times 10^{-14} \text{ m} = 45.5 \text{ fm}$

unit 1:

1.) $C = Q/V$

$$= \frac{3.00 \times 10^{-6} \text{ C}}{120 \text{ V}}$$

$$= \boxed{2.50 \times 10^{-8} \text{ F} = 25.0 \text{ nF}}$$

2.) a. $U = \frac{1}{2} (10 \times 10^{-6}) (9 \times 10^3)^2$

$$= \boxed{405 \text{ J}}$$

b. $V = \sqrt{\frac{2(40)}{8 \times 10^{-6}}}$

$$= \boxed{3.16 \times 10^3 \text{ V}}$$

3.) each capacitor = $\left(\frac{C_{\text{total}}}{4} = \frac{10.0 \times 10^{-6}}{4} = \boxed{2.50 \times 10^{-6} \text{ F}} \right)$

charge stored on each = $Q = (2.50 \times 10^{-6}) (9 \times 10^3) = \boxed{22.5 \text{ mC}}$

series reduces capacitance and limits total charge

4.) a.) $I = V/R$

$$= 3 \text{ V} / 1003 \Omega$$

$$= \boxed{2.99 \text{ mA}}$$

b.) $P = I^2 R$

$$= (2.99 \times 10^{-3})^2 (3)$$

$$= \boxed{2.68 \times 10^{-5} \text{ W}}$$

unit 2:

$$i.) \frac{1}{R_3} = \frac{1}{2000} - \frac{1}{3333.3}$$
$$= \frac{1}{0.0002} = 5000 \Omega$$

$$R_3 = \boxed{5k\Omega}$$

$$c. I_1 = \frac{V}{R_1} = \frac{12}{10000} = \boxed{1.2mA}$$

$$I_2 = \frac{12}{5000} = \boxed{2.4mA}$$

$$I_3 = \frac{12}{5000} = \boxed{2.4mA}$$

$$b. I = \Delta V / R_{tot}$$
$$= 12V / 2000 \Omega$$
$$= \boxed{6.00mA}$$

2.) a. no internal resistance

$$\text{series: } I = V/R = 3/500 = 6.00mA$$

$$P = IV = (6.00mA)(3) = \boxed{18mW}$$

$$\text{parallel: } I = 6.00mA$$

$$P = \boxed{18mW}$$

b. w/ internal resistance

$$\text{series: } R_{total} = 500 + 0.5 + 0.5 = 501 \Omega$$

$$I = 3/501 = 5.98mA$$

$$P = (5.98)^2 (501 \Omega) = \boxed{17.915mW}$$

$$\text{parallel: } R_{total} = 500.25 \Omega$$

$$I = 3/500.25 = 5.999mA$$

$$P = (5.999)^2 (500.25) = \boxed{18.007mW}$$

$$3.) a. I = \frac{V}{R} = 6.27V / 10 + .16 = \boxed{0.617A}$$

$$b. P = I^2 R = (0.617)^2 \cdot 10 = \boxed{3.81W}$$

$$c. I = \sqrt{.5/10} = 0.224A$$

$$R_{tot} = V/I = 6.27 / .224 = 28.04 \Omega - 10 \Omega = 18.04 \Omega$$
$$= 18.04 \Omega - 0.06 = \boxed{17.98 \Omega}$$

unit 3:

- 1.) a. $\hat{j}$ up $\hat{k}$ out $-\hat{i}$ left
b. $\hat{j}$ up $\hat{i}$ right $-\hat{k}$ into page
c. $\hat{i}$ right $-\hat{k}$ into page $\hat{j}$ down

2.) $\sin \theta = \frac{1.40 \times 10^{-16}}{(1.60 \times 10^{-19})(4 \times 10^3)(1.25)}$
 $= 0.175$

$\theta_1 = \sin^{-1}(0.175)$

$\theta_1 = 10.1^\circ$

$\theta_2 = 180^\circ - 10.1^\circ = 169.9^\circ$

3.) a. $q = mv / rB$
 $= \frac{(2.66 \times 10^{-26})(5 \times 10^6)}{0.231(1.20)}$
 $= 4.8 \times 10^{-19} \text{ C}$

b. $\frac{q}{e} = \frac{4.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 3$

c. 0.260 m